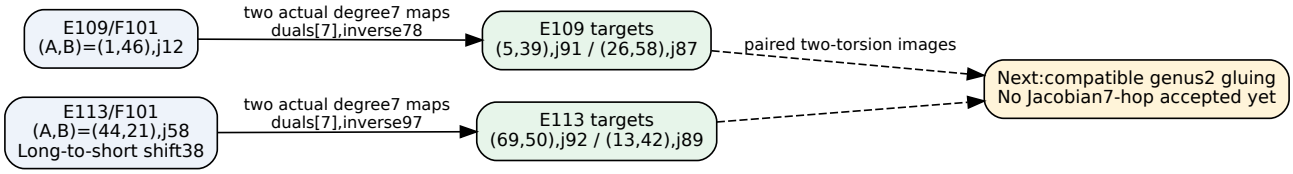
Certified degree7 maps for both elliptic factors

2026-11-11 · Chained ECDLP Transfers · CT19

Unsigned/unverified standing. Four actual F101 maps and their duals pass exact polynomial identities and complete109/113-point group controls. Another unchanged arithmetic/kernel engine replays the frozen certificates. These reuse known elliptic algorithms;no Jacobian7-hop or computational advantage is accepted yet. PCCa51d74/beeac7/45f608. [pcc:unverified]



Exact models,kernels and duals

Original E109 is $y^2=X^3+X+46$. The second quotient has long coefficients [5,33,13,1];X=x+38 gives short(A,B)=(44,21),order113. Every101^2 affine coordinate-bridge case is checked. [pcc:unverified]

E109 kernels[9,76,20,1]/[72,100,35,1] give(5,39)/(26,58). E113 kernels[59,46,8,1]/[77,86,71,1] give(69,50)/(13,42). Their source j-values12/58 and target pairs91/87,92/89 are distinct; checked coefficient-scale tests and the independent engine reject all source-target/cross-target model equivalences. [pcc:unverified]

For x-map N/D,$D=h^2$,put S=N'D−ND'. The exact F101[x] equality is $f(\text{source})\cdot S^2=D\cdot(N^3+A(\text{target})\cdot N\cdot D^2+B(\text{target})\cdot D^3)$. Smoothness,squarefree kernel,source-branch coprimality and coprime map numerator/denominator pass. The resulting nonconstant pointed map extends to a separable degree7 elliptic isogeny;that geometric bridge remains written. [pcc:unverified]

Actual duals normalize withu=1/7=29,$x\to u^2x$,$y\to u^3y$. Both global rational-x compositions equal multiplication[7] by independent division-polynomial checks. Every109/113 source/codomain point,bijection,all pair additions, scalars,signed dual norms and inverse78/97 recovery pass. Maps read coordinates,not unknown DLP labels. [pcc:unverified]

Independent replay,source custody and next geometry

The unchanged Montgomery field/poly/kernel engine imports no producer Kohel/new transfer code. It independently rechecks kernel membership/ subgroup closure,Vélu codomain,exact curve/global-x identities and complete point controls. Root example and the same-harness consumer integration pass; these are two entrypoints,not two independent scientific verdicts. [pcc:unverified]

TSV SHA25678f0b33e611113f8b6c8340874c39f153c0620b91a86c8242c606c675a927615 binds four forward/dual records. Separate manifests freeze102 construction and five verifier inputs at unchanged crypto75a737b8;seven native runtime guards remain distinct. CI checks out both closures and Cargo.lock freezes dependencies. No arithmetic engine or lint exemption changed. [pcc:unverified]

Four EllipticSeven.lean certificates pass gate4,PCC58dbb8:general cleared long-to-short coordinates,isomorphism scaling,and inverse7 recovery under 109/113 torsion. Total72 written lemmas. They do not formalize finite fields,kernels,elliptic geometry or a Jacobian7-map. [pcc:unverified]

Preserved corrections and exact test scope

Development attempt02 mislabelled an old helper’s unchecked scale candidates as verified isomorphisms. Actual coefficient equations/distinct j correct that interpretation in03;the independent engine agrees. The misleading diagnostic remains preserved. Initial adapter workspace/formatting errors were resolved;no engine code was edited. [pcc:unverified]

Targeted full maps/duals,independent replay,formatting,strict Clippy and all source guards pass. Current full native25/reference51-unit-plus2- integration CI pending;one old benchmark ignored. Prior75e5064d1 exact-head 43030710539 passes native25/reference50-unit-plus1-integration;PR1482 merged29c4d7206. Historical receipt scopes stay immutable. [pcc:unverified]

Run consumer check.sh with Rust1.99.1/pinned sibling sources. Root: `cargo test --release --example chained_elliptic_seven_reference`. Targeted consumer tests:`actual_seven_kernels_duals_global_identities_and_full_groups, seven_independent`. Source/receipts/ full account:`ELLIPTIC_SEVEN.md`. [pcc:unverified]

Genuine Jacobian continuation remains next

CT20 glues compatible images of the original two-torsion pairs. It must verify geometric kernel inclusion/global factorization,twists,descent, actual codomains,duals,full rational domains and order109 preservation before accepting a genuine Jacobian7-hop. Original-source and direct elliptic controls remain mandatory in any later cost comparison. [pcc:unverified]

These algorithms are existing Vélu/Kohel/division-polynomial methods. Kohel thesis primary retrieval failed and is not marked reviewed. Reviewed Milne geometric inputs,exact identities and independent replay support this bounded scope. No benchmark,solver gain,paid job,novelty or cross-prime descent. S3/live-budget,KB/vault and crypto frozen-ledger/archive gates persist. [pcc:unverified]